

Responses to reviewers

Reviewer's Responses to Questions, focusing on

- *Statistical significance and confidence*
- *Cross-model generality*
- *Replicability*
- *Supplementary materials*

Reviewer #1: The study's objectives and rationale are generally clear but could be more sharply **focused**.

R4Tun adopts an LLM-driven agentic system to an expert-designed tunnel segmentation pipeline to improve robustness under varying tunnel conditions without expert tuning.

The introduction and abstract adequately outline the need for adaptive tunnel inspection and the limitations of current methods, yet the specific research goals are somewhat dispersed. Please consider consolidating the objectives into a distinct subsection in the Introduction, and explicitly listing the **aims** of R4Tun more clearly.

R4Tun is designed to: (i) adapt stage parameters to current tunnel conditions, (ii) improve robustness under varying conditions, and (iii) clarify the contribution of memory, state, and knowledge within an LLM-driven reasoning framework.

Reviewer #3: The study's basic research objectives and rationale are presented, but they are not articulated with sufficient clarity and specificity to highlight the **novelty** of the work. While the goal of developing the SAM4Tun framework for tunnel point cloud segmentation is mentioned, the rationale for designing an inference-based multi-agent framework integrated with engineer-designed prompt-driven pipelines—especially the unique research gaps it addresses in tunnel detection with large language models (LLMs)—is not fully elaborated. This lack of in-depth articulation weakens the clarity of the study's core objectives and the motivation behind its design.

R4Tun's novelty lies in introducing and validating a context design based on memory, state, and knowledge for LLM-driven adaptation of an expert-designed tunnel segmentation pipeline.

The study is largely replicable due to the open-source code and dataset, and the method is described with reasonable detail. However, some implementation specifics are

omitted.

The prompt templates or system instructions applied in the method should be given in the **supplementary material**, which support a small, fully worked example of a CoT trace for a stage agent and the reflective agent.

We will add the following supplementary materials:

- *Stage-agent prompt templates*
- *Parameter set with descriptions*
- *Tunnel characteristics schema*
- *Stage intermediate state schema*
- *One full worked example (single tunnel, with full details)*

No, the methodological details are insufficient to ensure full replicability and reproducibility. Critical details are missing, including **the specific implementation logic of how inference agents delegate parameter tuning, the unreported pipeline parameter selection process of SAM4Tun** (which should be demonstrated via Chain-of-Thought (CoT)), and **the lack of ablation analyses to verify the contribution of core modules** (e.g., multi-agent collaboration, LLM-driven prompt tuning).

We addresses replicability through three additions:

- *A system architecture diagram and pseudocode show the full inference-to-parameter flow:*
 - *Context assembly → builds structured prompt per stage → sends to LLM → LLM returns JSON → JSON saved as parameters_{stage}.json → stage agent loads it and runs.*
- *A worked example demonstrates the 5-step CoT reasoning process:*
 - *Anchoring → classification → diagnostic inspection → parameter adaptation → self-validation → JSON output*
- *The context ablation (memory → +state → +knowledge) is the ablation verifying core modules, now with statistical evidence for each component's contribution.*

Additionally, **the limited LLM comparison and ambiguous performance descriptions** further reduce the transparency of the method, making it difficult for other researchers to replicate the study's experiments and results.

We conducted a comparison across the latest versions of 3 mainstream Large Language Models (LLMs) to validate cross-model performance.:

- *Claude Opus 4.6*
- *GPT 5.4*
- *Gemini 3.2*

We clarified performance descriptions by answering the following 4 questions:

- 1: Does context-driven adaptation help at all?
 - Yes. Our ablation shows overall mIoU improvement from 0.150 to [0.313 -- 0.328] across three LLMs, statistically significant ($p < 0.0001$) with effect sizes of $d = 1.4$ -- 1.9 , indicating that nearly every tunnel improved, not just a few.
- 2: Which context component drives the gain?
 - State. Our ablation shows that adding State produces the largest incremental gain: Memory+State improves mIoU by +0.149 to +0.162 across 3 LLMs (all $p < 0.0001$), accounting for the majority of the total improvement.
- 3. Does this hold across tunnel families?
 - Yes. The regular family showed an improvement from 0.367 to [0.586–0.621] with effect sizes $d = 1.4$ – 2.2 . The complex family showed an improvement from 0.042 to [0.159–0.177] with effect sizes $d = 1.2$ – 2.4 .
- 4. Does this hold across LLMs?
 - Yes. Our cross-model comparison shows all 3 LLMs produce overlapping 95% CIs on the m+s+k improvement, ranging from +0.121 to +0.219, indicating no statistically distinguishable difference across LLMs:
 - Claude Opus 4.6: [+0.137, +0.219]
 - OpenAI GPT-5.4: [+0.138, +0.204]
 - Google Gemini 3.2: [+0.121, +0.205]

(Each interval means that, with 95% confidence, the true mean mIoU improvement for that LLM lies between its lower and upper bound.)

Reviewer #1: Due to the **small sample size**, increase the sample size if possible, or employ cross-validation to strengthen generalizability.

We extended our sample size to 30 tunnel subsets:

- 13 regular (10 alternated regular and 3 continuous)
- 17 complex subsets characterised by a larger diameter (7.5 m vs 5.5 m), an additional segment per ring (7 vs 6), and no predictable K-block positional pattern.

Reviewer #3: Statistical analyses and reporting are incomplete and lack rigor. While basic performance metrics (e.g., mIoU) are reported, the study does not include key statistical indicators such as **P-values, confidence intervals (CIs), or effect sizes** to validate the significance of performance improvements—especially for the mIoU change in off-reference cases.

1. We added P-values and effective sizes to the overall mIoU:

- Overall mIoU improves from 0.150 to 0.313–0.328 across 3 LLMs, with $p < 0.0001$ and effect sizes $d = 1.4–1.9$, meaning nearly every tunnel improved with statistical significance.

2. We added cross-model 95% CI comparisons on the mIoU improvement:

- Opus 4.6: [+0.137, +0.219]
- GPT 5.4: [+0.138, +0.204]
- Gemini 3.2: [+0.121, +0.205]

All 3 intervals overlap, confirming that the improvement is driven by the context design, not the choice of model.

The experimental controls are also inadequate: the LLM comparison is limited to only two models with no unified control of variables (e.g., prompt templates, reasoning steps), and **the sampling mechanism for the Seg2Tunnel dataset** (e.g., stratification by tunnel scenario, point cloud characteristics) is not clearly described, undermining the credibility of the statistical results.

We separated the Seg2Tunnel dataset into two groups:

Regular tunnels (T1: 5, T2: 5, T3: only have 3 for sampling; n=13):

- Same diameter (5.5 m) and segment count (6) as the reference tunnel
- K-block position follows a predictable pattern
- Scanner centred or multi-station for manageable density distribution
- Pipeline was designed for this geometry

Complex tunnels (T4: 10, T5: only have 7 for sampling; n=17):

- 36% larger diameter (7.5 m), 50% longer rings (1.8 m vs 1.2 m), one extra segment per ring (7 vs 6)
- K-block position has no repeating pattern
- Scanner offset from tunnel axis indicates severe near/far density imbalance
- Every pipeline assumption is outside its design envelope

Reviewer #1: Consider adding **a comprehensive flow** to make the agent's operation more immediately understandable.

We removed the reflection agent, and the current flow is as follows:

Point cloud → Unfolding agent → Denoising agent → Enhancement agent → Prompting agent → SAM Segmentation → 3D reprojection"

Reviewer #3: The manuscript would benefit significantly from **adding key figures/tables** and optimizing the placement and presentation of existing ones. First, additional visualizations are needed: CoT-based figures demonstrating SAM4Tun's pipeline parameter selection (and comparison with R4Tun), ablation curves for core hyperparameters (e.g., number of inference agents, LLM reasoning steps), and a comparative table of multiple representative LLMs (including accuracy, reasoning efficiency, and robustness metrics). Second, the existing charts suffer from inappropriate placement that disrupts logical flow; these need to be reorganized to follow the **"figure follows text" principle**. No existing figures/tables require removal, but all need structural and contextual optimization for clarity.

We will provide:

Figures:

- *System architecture*
- *Context design (Memory (m), State (s), Knowledge (k))*
- *Agent architecture (m+s+k)*
- *Step-wise ablation mIoU performance split by regular vs complex (Barchart)*
 - *baseline →*
 - *m →*
 - *m+s →*
 - *m+s+k*
- *Cross-model average mIoU comparison (Line chart)*

Tables:

- *Dataset properties (regular and complex)*
- *Main ablation results — Rows: baseline, m, m+s, m+s+k. Columns per family: mean mIoU, delta, p, d.*
- *Cross-model summary — Rows: Gemini, GPT, Opus. Columns: m+s+k delta, 95% CI, d.*

Reviewer #1: Besides, **the claimed "transparency" and "engineer-guided"** nature are not validated with user studies. It remains unproven whether engineers can or will effectively use the logged traces to oversee or correct the system.

The overly general claims of ‘transparency’ are removed because we plan to do on-field user studies in the future with domain engineers.

The claim of ‘engineer-guided’ is replaced with a more specific claim as ‘reference parameter values extracted from SAM4Tun open-source code’.

Reviewer #3: The authors have mentioned some general **limitations** of the study, but the statement is not sufficiently clear, specific, or well-integrated with the research design and experimental results.

Limitations:

- *The study uses a single reference parameter set as the starting point for all adaptation.*
- *The study does not test generalisation beyond the Seg2Tunnel dataset.*
- *The study adapts parameters in a single forward pass without iterative self-correction.*

Reviewer #2:

The manuscript proposes a reasoning-based multi-agent framework for segmental tunnel analysis in point cloud data. The topic fits the scope of the journal. The idea is interesting. However, the current investigation is not sufficient to support the claimed robustness and methodological contribution. The following specific comments should be addressed before publication.

1. The authors should ~~provide the complete parameter set~~ (*We will provide it in the supplementary materials.*) for each stage, including default values and bounds. The state, knowledge, and memory should be clearly defined in an implementable way. The LLM configuration should also be reported, such as model version, decoding settings, and failure handling.
2. In the current version, the authors ~~only conducted the static comparison under the off-reference subsets.~~ (*The samples have now been compared against both the regular and complex families across a dataset comprising 30 tunnels, employing 3 LLMs.*) mIoU is improved by R4Tun based on the adaptive parameters, but it does not isolate the contribution of the reasoning enabled LLM acting as a white box controller. Table 2 suggests that state updating provides the largest gain. Therefore the paper needs stronger evidence that the improvement is specifically due to the LLM-based reasoning.
3. The paper defines knowledge as concise parameter definitions and tuning rules based on expert-written tuning guidelines. However, no deterministic execution of these tuning rules is provided as a baseline. It is ambiguous how much of the reported gain comes from the rules themselves or the LLM generated updates. The experiment that ~~applies the same tuning rules without LLM inference~~ (*we ran SAM4Tun’s fixed parameters on all 30 tunnels as baselines*) regular and complex) is required to support the claimed

role of the LLM based agents.

4. mIoU is a relevant but not sufficient factor to assess the performance of the proposed framework. The authors should report ~~IoU of each class~~ (*We will provide it in the supplementary materials.*) and provide stronger evidence on boundary quality near joints and segment edges. The paper should also report ~~runtime, number of calls of LLM~~ (*We will provide them in the experiment result section.*) and sensitivity to the reflection step, because these factors affect practical applications.

5. Most figures are blurred. High-quality figures (not yet started) are required, and figure labels should be readable at column width.

Dear all,

The paper was rejected, but luckily, the problems all came from writing, figures, and experiments. I think all of them can be resolved with more time. I list some thoughts for the comments:

1. Are the objectives and the rationale of the study clearly stated?

Reviewer #1: The study's objectives and rationale are generally clear but could be more sharply focused. The introduction and abstract adequately outline the need for adaptive tunnel inspection and the limitations of current methods, yet the specific research goals are somewhat dispersed. Please consider consolidating the objectives into a distinct subsection in the Introduction, and explicitly listing the aims of R4Tun more clearly.

Easy to solve by editing the writing.

Reviewer #2: No

Reviewer #3: The study's basic research objectives and rationale are presented, but they are not articulated with sufficient clarity and specificity to highlight the novelty of the work. While the goal of developing the SAM4Tun framework for tunnel point cloud segmentation is mentioned, the rationale for designing an inference-based multi-agent framework integrated with engineer-designed prompt-driven pipelines—especially the unique research gaps it addresses in tunnel detection with large language models (LLMs)—is not fully elaborated. This lack of in-depth articulation weakens the clarity of the study's core objectives and the motivation behind its design.

As discussed with Tao, there all few research on a multi-agent framework for tunnel, we need to edit the writing to show this and the objectives and the motivation with the multi-agent framework.

2. If applicable, is the application/theory/method/study reported in sufficient detail to allow for its replicability and/or reproducibility?

Reviewer #1: Mark as appropriate with an X:

Yes ☐ No ☒ N/A ☐

Provide further comments here:

The study is largely replicable due to the open-source code and dataset, and the method is described with reasonable detail. However, some implementation specifics are omitted.

The prompt templates or system instructions applied in the method should be given in the supplementary material, which support a small, fully worked example of a CoT trace for a stage agent and the reflective agent.

Easy to add.

Reviewer #2: Mark as appropriate with an X:

Yes ☐ No ☒ N/A ☐

Provide further comments here:

Reviewer #3: Mark as appropriate with an X:

Yes ☐ No ☒ N/A ☐

Provide further comments here:

No, the methodological details are insufficient to ensure full replicability and reproducibility. Critical details are missing, including the specific implementation logic of how inference agents delegate parameter tuning, the unreported pipeline parameter selection process of SAM4Tun (which should be demonstrated via Chain-of-Thought (CoT)), and the lack of ablation analyses to verify the contribution of core modules (e.g., multi-agent collaboration, LLM-driven prompt tuning). Additionally, the limited LLM comparison and ambiguous performance descriptions further reduce the transparency of the method, making it difficult for other researchers to replicate the study's experiments and results.

Need more experiments, but it can be solved.

3. If applicable, are statistical analyses, controls, sampling mechanism, and statistical reporting (e.g., P-values, CIs, effect sizes) appropriate and well described?

Reviewer #1: Due to the small sample size, increase the sample size if possible, or employ cross-validation to strengthen generalizability.

Need more experiments, but it can be solved.

Reviewer #2: No

Reviewer #3: Statistical analyses and reporting are incomplete and lack rigor. While basic performance metrics (e.g., mIoU) are reported, the study does not include key statistical indicators such as P-values, confidence intervals (CIs), or effect sizes to validate the significance of performance improvements—especially for the mIoU change in off-reference cases. The experimental controls are also inadequate: the LLM comparison is limited to only two models with no unified control of variables (e.g., prompt templates, reasoning steps), and the sampling mechanism for the Seg2Tunnel dataset (e.g., stratification by tunnel scenario, point cloud characteristics) is not clearly described, undermining the credibility of the statistical results.

Need more experiments, but it can be solved.

4. Could the manuscript benefit from additional tables or figures, or from improving or removing (some of the) existing ones?

Please provide specific suggestions for improvements, removals, or additions of figures or tables. Please number each suggestion so that author(s) can more easily respond.

Reviewer #1: Consider adding a comprehensive flow to make the agent's operation more immediately understandable.

Easy to solve by adding one paragraph or subsection about the comprehensive flow.

Reviewer #2: No

Reviewer #3: The manuscript would benefit significantly from adding key figures/tables and optimizing the placement and presentation of existing ones. First, additional visualizations are needed: CoT-based figures demonstrating SAM4Tun's pipeline parameter selection (and comparison with R4Tun), ablation curves for core hyperparameters (e.g., number of inference agents, LLM reasoning steps), and a comparative table of multiple representative LLMs (including accuracy, reasoning efficiency, and robustness metrics). Second, the existing charts suffer from inappropriate placement that disrupts logical flow; these need to be reorganized to follow the "figure follows text" principle. No existing figures/tables require removal, but all need structural and contextual optimization for clarity.

Need more visualizations and refine current visualizations.

5. If applicable, are the interpretation of results and study conclusions supported by the data?

Reviewer #1: Mark as appropriate with an X:

Yes ☒ No ☐ N/A ☐

Provide further comments here:

N/A

Reviewer #2: Mark as appropriate with an X:

Yes ☐ No ☒ N/A ☐

Provide further comments here:

Reviewer #3: Mark as appropriate with an X:

Yes ☒ No ☐ N/A ☐

Provide further comments here:

The interpretation of results and study conclusions are partially supported by basic experimental data but lack sufficient and rigorous evidence to validate their claims. While the manuscript reports SAM4Tun's performance on the Seg2Tunnel dataset, the limited LLM comparisons, missing ablation analyses, and ambiguous mIoU performance statements (in off-reference cases) mean the conclusions about the framework's superiority, adaptability, and ability to prevent baseline collapse are not fully substantiated. Additionally, the lack of interpretable analyses (e.g., CoT parameter comparison) prevents a deep interpretation of why the framework performs well, leading to superficial result interpretation that is not fully aligned with comprehensive data support.

Need more experiments and writing edition.

6. Have the authors clearly emphasized the strengths of their study/theory/methods/argument?

Reviewer #1: Yes.

Reviewer #2: No

Reviewer #3: No, the authors have not clearly or sufficiently emphasized the study's strengths. While the SAM4Tun framework integrates innovative design elements (e.g., inference-based multi-agent collaboration, engineer-designed prompt-driven pipelines for tunnel point cloud segmentation), the manuscript fails to explicitly highlight these unique strengths—such as the framework's ability to address LLM-driven point cloud segmentation challenges in tunnel detection, its advantages in off-reference scenario robustness, or the practical value of its intelligent visualization and parameter tuning. The strengths are only implied rather than

systematically articulated and contrasted with state-of-the-art methods, making the study's contributions unremarkable.

Need writing edition.

7. Have the authors clearly stated the limitations of their study/theory/methods/argument?

Reviewer #1: The study uses only five tunnel subsets for evaluation. This small sample size restricts statistical power and the ability to generalize the findings across the full diversity of real-world tunnel conditions.

Besides, the claimed "transparency" and "engineer-guided" nature are not validated with user studies. It remains unproven whether engineers can or will effectively use the logged traces to oversee or correct the system.

Maybe delete "transparency" and "engineer-guided" nature? As it needs more identification with user studies and other strengths on parameter tuning and generalization are enough.

Reviewer #2: No

Reviewer #3: The authors have mentioned some general limitations of the study, but the statement is not sufficiently clear, specific, or well-integrated with the research design and experimental results.

Need writing edition.

8. Does the manuscript structure, flow or writing need improving (e.g., the addition of subheadings, shortening of text, reorganization of sections, or moving details from one section to another)?

Reviewer #1: Please consider introducing the overall R4Tun workflow (Fig. 2) and agent architecture (Fig. 3) before diving into the detailed CoT explanation. Move the detailed mathematical formulation (Eq. 1) and specific agent operations (Sec. 2.3) to a sub-section after the core concepts are established.

Need writing edition.

Reviewer #2: Yes

Reviewer #3: Yes, the manuscript's structure, flow, and writing require substantial improvement. First, the logical flow is disrupted by inappropriate chart placement, which disconnects visual evidence from textual analysis. Second, key sections lack hierarchical subheadings (e.g., for the CoT parameter selection process, ablation

analysis, and LLM comparison experiments), leading to disorganized presentation of critical content. Third, there is redundant and ambiguous writing—such as the repeated, semantically unclear statement about mIoU in off-reference cases—that needs to be revised for conciseness and accuracy. Additionally, the discussion section needs reorganization to better link experimental results to conclusions, and methodological details (e.g., agent parameter tuning) should be moved to a dedicated subsection for clarity and readability.

Need writing edition.

9. Could the manuscript benefit from language editing?

Reviewer #1: Yes

Reviewer #2: Yes

Reviewer #3: Yes

Associate Editor: Thank you for considering Automation in Construction for the publication of your manuscript. The reviewers have evaluated your work and based on those reviews, the work unfortunately needs to be rejected at this point. Several strong points are made by the reviewers. Please consider the provided reviewer comments for helping you to improve your work. [After significant revision, the authors of course still have the option to resubmit their work. In that case, no track changes need to be included, but the cover letter \(1-2 pages\) preferably indicates which changes have been made compared to the current version.](#)

In addition to addressing the reviewer comments, the authors could also do the following to improve the article quality:

- The journal uses a consecutive numbered reference style, starting at number [1]. Please review the author guide and try to comply from the first submission.
- Aim to include a dedicated Related Works section in Section 2. Do not limit yourself to add related works only briefly in the Introduction. This is typically insufficient to show the added contributions of the article.
- The article uses quite large page-wide figures. As a result, references to those Figures come several pages earlier. The authors may want to try to use the journal template in 2 columns, and see if they can fit the figures in the columns. In case of too many Figures, a solution might be to reduce the number of figures and/or use the Appendix space to include Figures and further details.
- Section 3 is typically included in the Results section (Section 4).
- The supplementary material can be included directly in the article, after the references, as regular appendices.

Need some writing edition for AE's comments to improve the article quality.

Reviewer #2:

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2. In the current version, the authors only conducted the static comparison under the off-reference subsets. mIoU is improved by R4Tun based on the adaptive parameters, but it does not isolate the contribution of the reasoning enabled LLM acting as a white box controller. Table 2 suggests that state updating provides the largest gain. Therefore the paper needs stronger evidence that the improvement is specifically due to the LLM-based reasoning.
3. The paper defines knowledge as concise parameter definitions and tuning rules based on expert-written tuning guidelines. However, no deterministic execution of these tuning rules is provided as a baseline. It is ambiguous how much of the reported gain comes from the rules themselves or the LLM generated updates. The experiment that applies the same tuning rules without LLM inference is required to support the claimed role of the LLM based agents.
4. mIoU is a relevant but not sufficient factor to assess the performance of the proposed framework. The authors should report IoU of each class and provide stronger evidence on boundary quality near joints and segment edges. The paper should also report runtime, number of calls of LLM and sensitivity to the reflection step, because these factors affect practical applications.
5. Most figures are blurred. High-quality figures are required, and figure labels should be readable at column width.

Overall, these are all related to writing, supplementary to experiment details, more experiments, and ablation studies.

Reviewer #3: No further comment

The AE of AIC suggests we can resubmit after resolving these comments. In addition, Elsevier also give 5 suggestions to transfer to other journals (attached PDF). Do you have any suggestions on this? Keep resubmitting to AIC or resubmit to another suggested journal?

Best regards,
Guangming